Week3_NYPD_Project

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```r
knitr::opts_chunk$set(echo = TRUE, message = FALSE, warning = FALSE)

library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.1     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.2.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(lubridate)
library(ggplot2)
```

```r
#Loading Data

url_NYPD <- "https://data.cityofnewyork.us/api/views/833y-fsy8/rows.csv?accessType=DOWNLOAD"

NYPD_raw <- read_csv(url_NYPD, show_col_types = FALSE)

glimpse(NYPD_raw)
```

```
## Rows: 29,744
## Columns: 21
## $ INCIDENT_KEY            <dbl> 231974218, 177934247, 255028563, 25384540, 726~
## $ OCCUR_DATE              <chr> "08/09/2021", "04/07/2018", "12/02/2022", "11/~
## $ OCCUR_TIME              <time> 01:06:00, 19:48:00, 22:57:00, 01:50:00, 01:58~
## $ BORO                    <chr> "BRONX", "BROOKLYN", "BRONX", "BROOKLYN", "BRO~
## $ LOC_OF_OCCUR_DESC       <chr> NA, NA, "OUTSIDE", NA, NA, NA, NA, NA, NA, NA,~
## $ PRECINCT                <dbl> 40, 79, 47, 66, 46, 42, 71, 69, 75, 69, 40, 42~
## $ JURISDICTION_CODE       <dbl> 0, 0, 0, 0, 0, 2, 0, 2, 0, 0, 0, 2, 0, 0, 2, 0~
## $ LOC_CLASSFCTN_DESC      <chr> NA, NA, "STREET", NA, NA, NA, NA, NA, NA, NA, ~
## $ LOCATION_DESC           <chr> NA, NA, "GROCERY/BODEGA", "PVT HOUSE", "MULTI ~
## $ STATISTICAL_MURDER_FLAG <lgl> FALSE, TRUE, FALSE, TRUE, TRUE, FALSE, TRUE, F~
## $ PERP_AGE_GROUP          <chr> NA, "25-44", "(null)", "UNKNOWN", "25-44", "18~
## $ PERP_SEX                <chr> NA, "M", "(null)", "U", "M", "M", NA, NA, "M",~
```

```
## $ PERP_RACE             <chr> NA, "WHITE HISPANIC", "(null)", "UNKNOWN", "BL~
## $ VIC_AGE_GROUP         <chr> "18-24", "25-44", "25-44", "18-24", "<18", "18~
## $ VIC_SEX               <chr> "M", "M", "M", "M", "F", "M", "M", "M", "M", "~
## $ VIC_RACE              <chr> "BLACK", "BLACK", "BLACK", "BLACK", "BLACK", "~
## $ X_COORD_CD            <dbl> 1006343.0, 1000082.9, 1020691.0, 985107.3, 100~
## $ Y_COORD_CD            <dbl> 234270.0, 189064.7, 257125.0, 173349.8, 247502~
## $ Latitude              <dbl> 40.80967, 40.68561, 40.87235, 40.64249, 40.845~
## $ Longitude             <dbl> -73.92019, -73.94291, -73.86823, -73.99691, -7~
## $ Lon_Lat               <chr> "POINT (-73.92019278899994 40.80967347200004)"~
```

```
#Cleaning Data

NYPD_clean <- NYPD_raw %>%
  select(
    OCCUR_DATE, OCCUR_TIME, BORO, PRECINCT,
    STATISTICAL_MURDER_FLAG, VIC_AGE_GROUP, VIC_SEX, VIC_RACE
  ) %>%
  mutate(
    OCCUR_DATE = mdy(OCCUR_DATE),
    OCCUR_TIME = hms(OCCUR_TIME),
    Year       = year(OCCUR_DATE),
    Month      = month(OCCUR_DATE, label = TRUE, abbr = TRUE),
    Dow        = wday(OCCUR_DATE, label = TRUE, abbr = TRUE),
    Hour       = hour(OCCUR_TIME),
    STATISTICAL_MURDER_FLAG = as.logical(STATISTICAL_MURDER_FLAG),
    Shootings  = 1
  )

summary(NYPD_clean)
```

```
##    OCCUR_DATE          OCCUR_TIME                           BORO
##  Min.   :2006-01-01   Min.   :0S                          Length:29744
##  1st Qu.:2009-10-29   1st Qu.:3H 30M 45S                  Class :character
##  Median :2014-03-25   Median :15H 15M 0S                  Mode  :character
##  Mean   :2014-10-31   Mean   :12H 46M 10.8747982786153S
##  3rd Qu.:2020-06-29   3rd Qu.:20H 44M 0S
##  Max.   :2024-12-31   Max.   :23H 59M 0S
##
##     PRECINCT      STATISTICAL_MURDER_FLAG VIC_AGE_GROUP        VIC_SEX
##  Min.   :  1.00   Mode :logical           Length:29744       Length:29744
##  1st Qu.: 44.00   FALSE:23979             Class :character   Class :character
##  Median : 67.00   TRUE :5765              Mode  :character   Mode  :character
##  Mean   : 65.23
##  3rd Qu.: 81.00
##  Max.   :123.00
##
##    VIC_RACE              Year          Month          Dow           Hour
##  Length:29744       Min.   :2006   Jul     : 3513   Sun:5872   Min.   : 0.0
##  Class :character   1st Qu.:2009   Aug     : 3352   Mon:4274   1st Qu.: 3.0
##  Mode  :character   Median :2014   Jun     : 3091   Tue:3469   Median :15.0
##                     Mean   :2014   Sep     : 2808   Wed:3283   Mean   :12.3
##                     3rd Qu.:2020   May     : 2795   Thu:3295   3rd Qu.:20.0
##                     Max.   :2024   Oct     : 2483   Fri:3901   Max.   :23.0
##                                    (Other):11702    Sat:5650
```

```
##    Shootings
##   Min.   :1
##   1st Qu.:1
##   Median :1
##   Mean   :1
##   3rd Qu.:1
##   Max.   :1
##
```

```
#Missing values
NYPD_clean %>%
  summarize(across(everything(), ~ sum(is.na(.)))) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "n_missing") %>%
  arrange(desc(n_missing))
```

```
## # A tibble: 13 x 2
##    variable                n_missing
##    <chr>                       <int>
##  1 OCCUR_DATE                      0
##  2 OCCUR_TIME                      0
##  3 BORO                            0
##  4 PRECINCT                        0
##  5 STATISTICAL_MURDER_FLAG         0
##  6 VIC_AGE_GROUP                   0
##  7 VIC_SEX                         0
##  8 VIC_RACE                        0
##  9 Year                            0
## 10 Month                           0
## 11 Dow                             0
## 12 Hour                            0
## 13 Shootings                       0
```

```
#Figure 1 - Shootings by Borough
NYPD_clean %>%
  ggplot(aes(BORO, fill = BORO)) +
  geom_bar() +
  labs(
    title = "NYPD Shooting Incidents by Borough",
    subtitle = "2006-2024",
    x = "NYC Boroughs",
    y = "Total Shootings",
    caption = "Figure 1 - Shootings by Borough"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

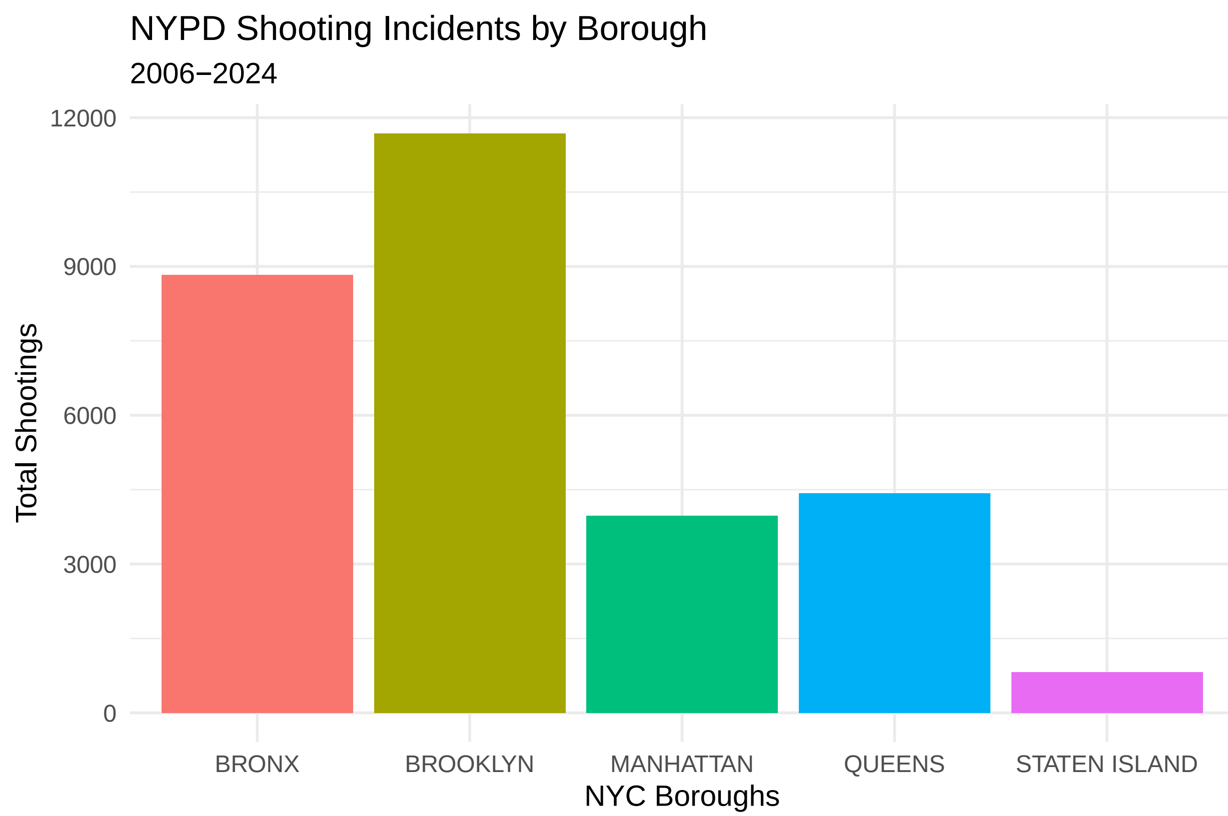

Figure 1 − Shootings by Borough

```r
#Figure 2 - Shootings by Year
NYPD_clean %>%
  ggplot(aes(Year)) +
  geom_bar(fill = "blue") +
  labs(
    title = "NYPD Shooting Incidents by Year",
    subtitle = "2006-2024",
    x = "Year",
    y = "Total Shootings",
    caption = "Figure 2 - Shootings by Year"
  ) +
  theme_minimal()
```

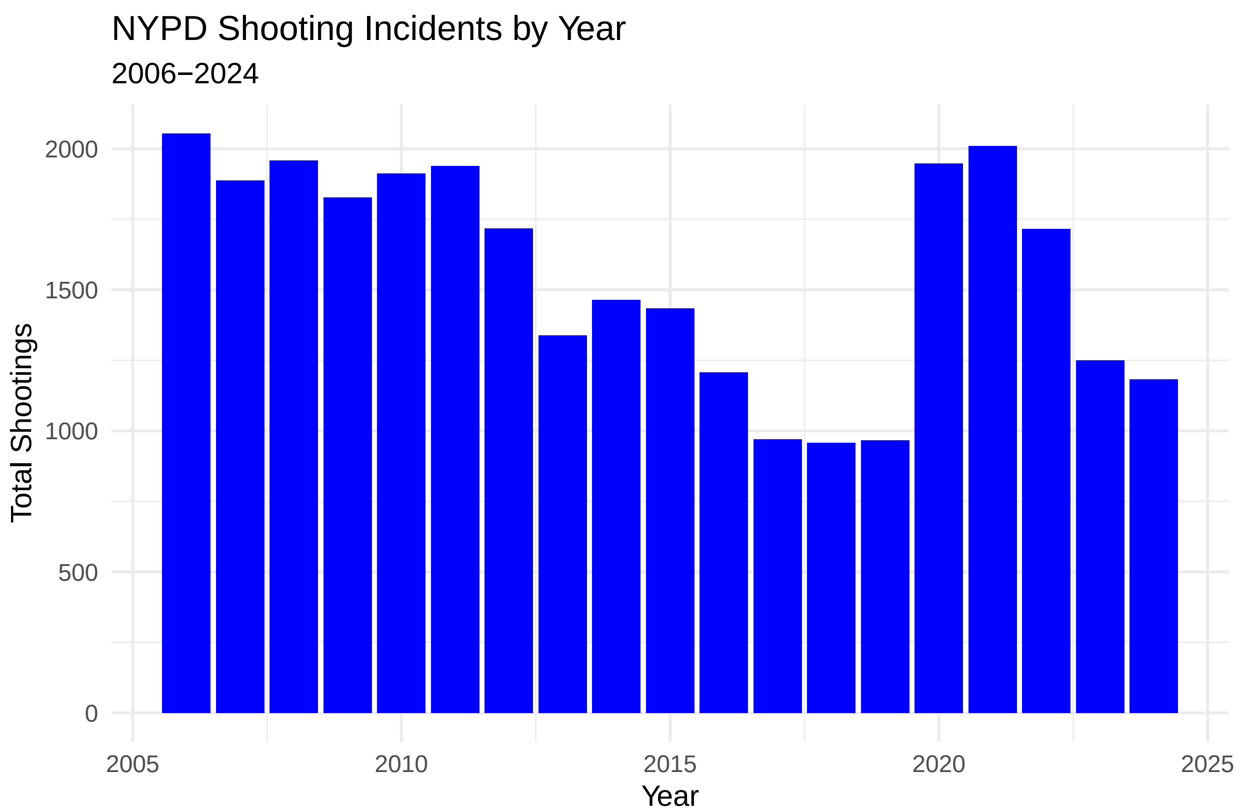

Figure 2 − Shootings by Year

```r
# Yearly summary data

NYPD_year <- NYPD_clean %>%
  group_by(Year) %>%
  summarize(
    Shootings = sum(Shootings),
    Murders = sum(STATISTICAL_MURDER_FLAG),
    .groups = "drop"
  )

NYPD_year %>% slice_max(Shootings, n = 4)
```

```
## # A tibble: 4 x 3
##    Year Shootings Murders
##   <dbl>     <dbl>   <int>
## 1  2006      2055     445
## 2  2021      2011     428
## 3  2008      1959     362
## 4  2020      1948     366
```

```r
NYPD_year %>% slice_min(Shootings, n = 4)
```

```
## # A tibble: 4 x 3
##    Year Shootings Murders
##   <dbl>     <dbl>   <int>
```

```
## 1  2018       958     204
## 2  2019       967     184
## 3  2017       970     174
## 4  2024      1182     239
```

```r
#Yearly Trend

NYPD_year %>%
  ggplot(aes(Year, Shootings)) +
  geom_line() +
  geom_point(color = "blue") +
  labs(
    title = "NYPD Shooting Incidents by Year",
    subtitle = "2006-2024",
    x = "Year",
    y = "Total Shootings",
    caption = "Figure 3-Yearly Trend"
  ) +
  theme_minimal()
```

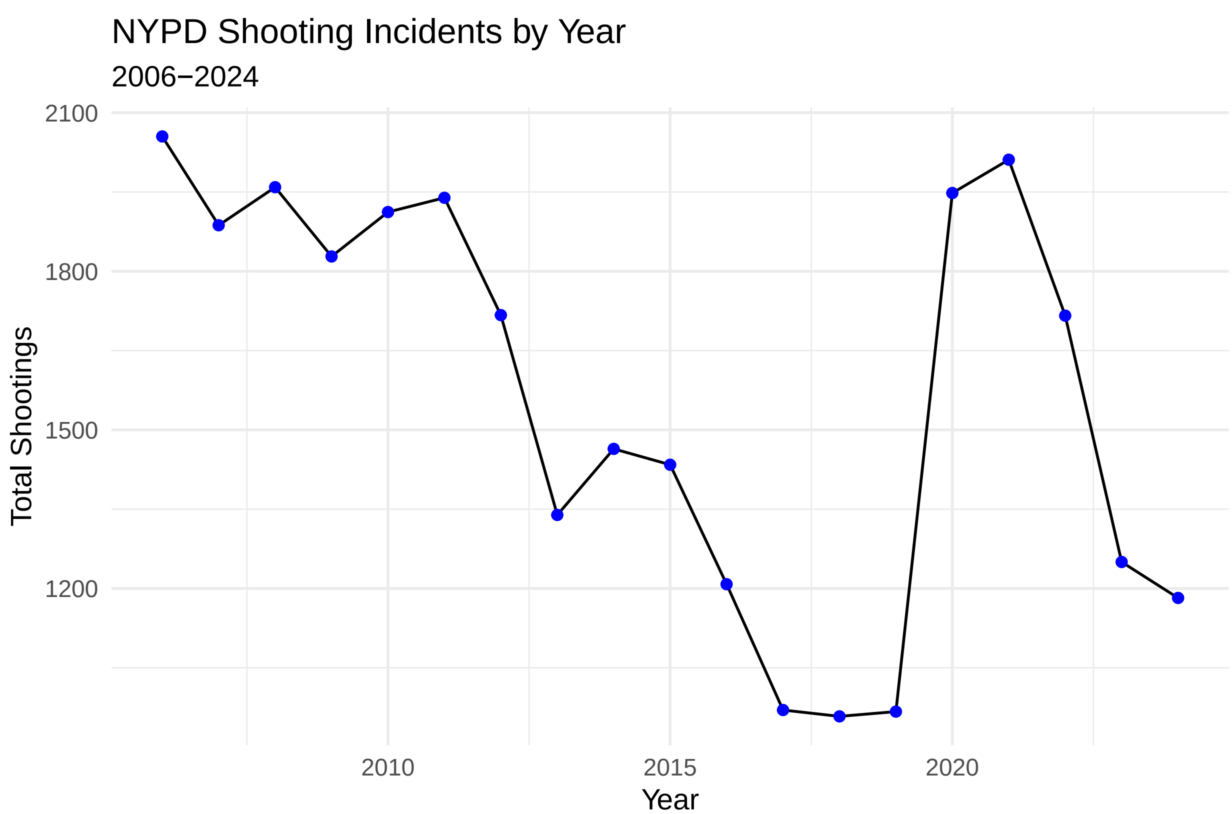

Figure 3−Yearly Trend

```r
# Bourough yearly summary

NYPD_boro_year <- NYPD_clean %>%
  group_by(BORO, Year) %>%
  summarize(
```

```r
    Shootings = sum(Shootings),
    Murders = sum(STATISTICAL_MURDER_FLAG),
    .groups = "drop"
  )

NYPD_boro_total <- NYPD_boro_year %>%
  group_by(BORO) %>%
  summarize(Shootings = sum(Shootings))
```

```r
NYPD_boro_year %>%
  ggplot(aes(Year, Shootings, color = BORO)) +
  geom_line() +
  geom_point() +
  labs(
    title = "NYPD Shootings by Borough Over Time",
    subtitle = "2006-2024",
    x = "Year",
    y = "Shootings",
    caption = "Figure 4 - Shootings by Borough Over Time"
  ) +
  theme_minimal()
```

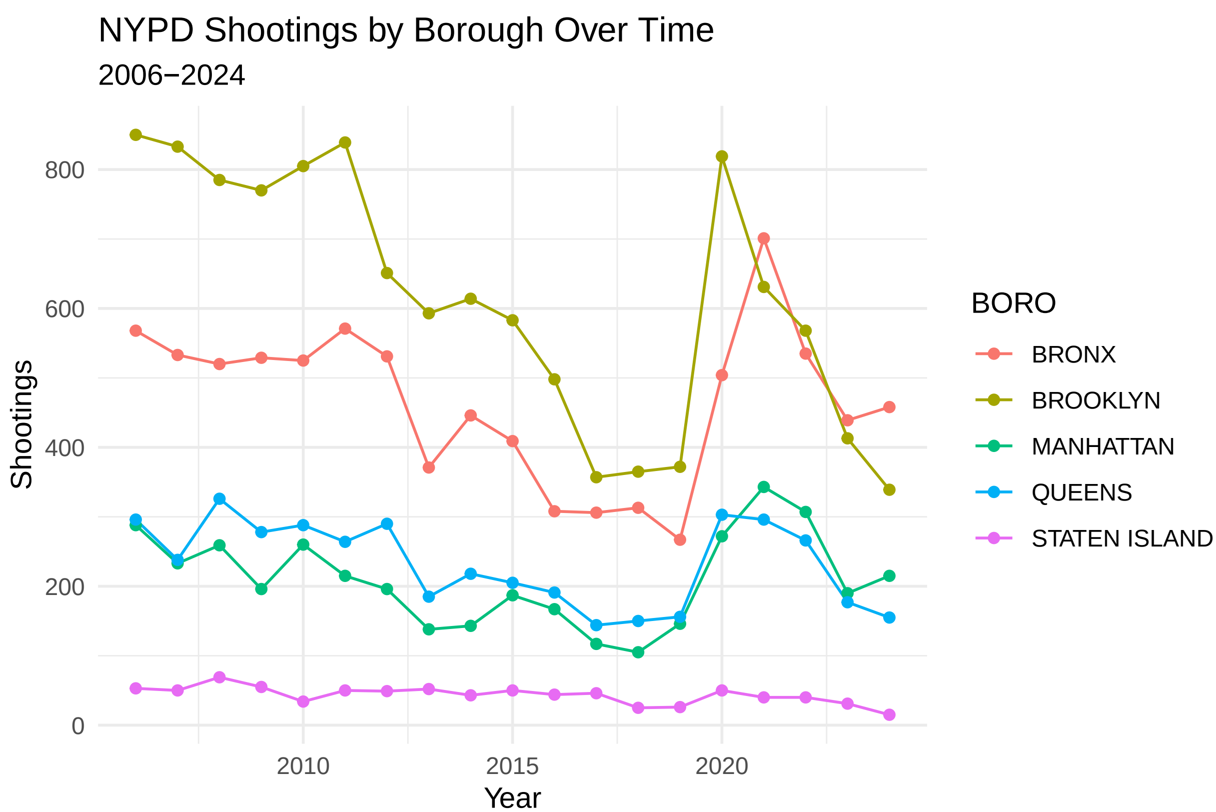

Figure 4 − Shootings by Borough Over Time

```r
#Daily Shootings
NYPD_date <- NYPD_clean %>%
  group_by(OCCUR_DATE) %>%
```

```r
  summarize(
    Shootings = sum(Shootings),
    Murders = sum(STATISTICAL_MURDER_FLAG),
    .groups = "drop"
  )

NYPD_date %>% slice_max(Shootings, n = 2)
```

```
## # A tibble: 2 x 3
##   OCCUR_DATE Shootings Murders
##   <date>         <dbl>   <int>
## 1 2020-07-05        47      11
## 2 2011-09-04        31       4
```

```r
#Shootings Per Day
NYPD_date %>%
  ggplot(aes(OCCUR_DATE, Shootings)) +
  geom_line() +
  scale_x_date(date_labels = "%Y %b") +
  labs(
    title = "NYPD Shootings Per Day",
    subtitle = "2006-2024",
    x = "Date",
    y = "Shootings",
    caption = "Figure 5 - Shootings Per Day"
  ) +
  theme_minimal()
```

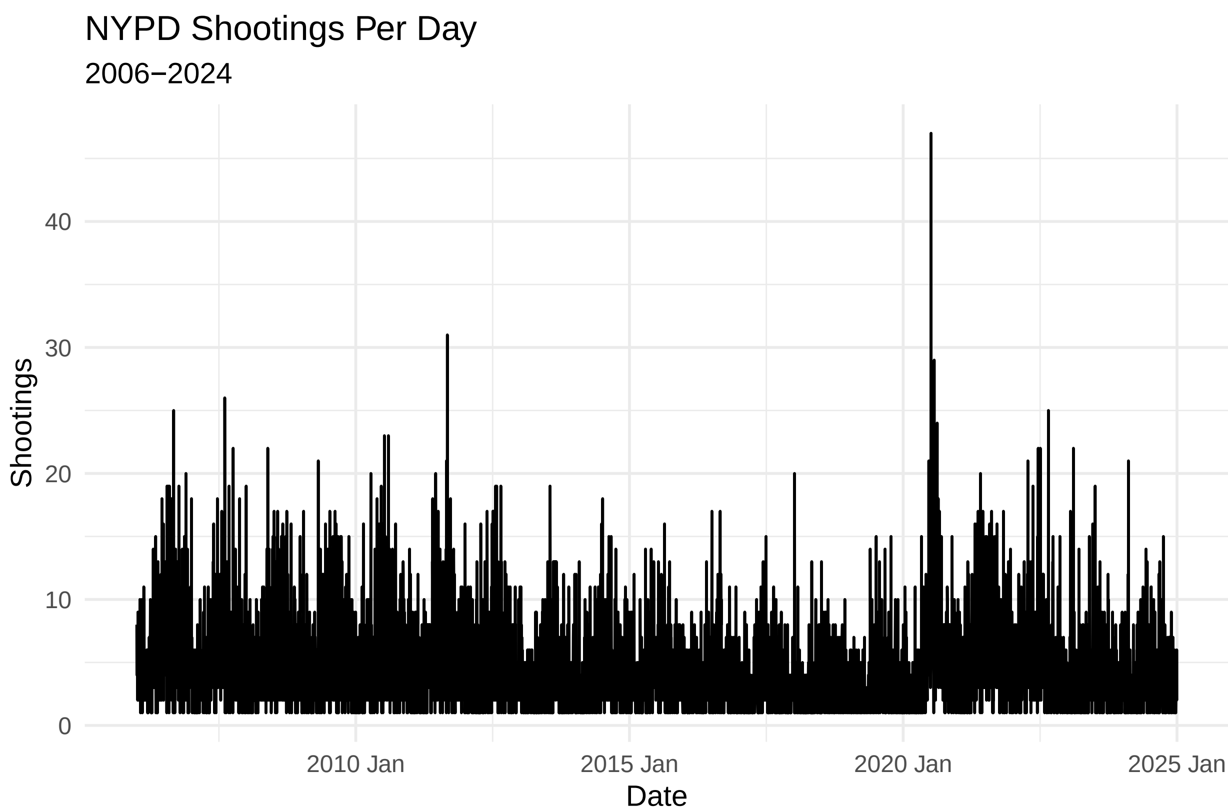

Figure 5 − Shootings Per Day

```r
#Shootings by Day of Year
NYPD_time_year <- NYPD_clean %>%
  mutate(DayOfYear = as.Date(format(OCCUR_DATE, "%m-%d"), "%m-%d")) %>%
  group_by(DayOfYear) %>%
  summarize(
    Shootings = sum(Shootings),
    Murders = sum(STATISTICAL_MURDER_FLAG),
    .groups = "drop"
  )
```

```r
#Shootings by Day of Year
NYPD_time_year %>%
  ggplot(aes(DayOfYear, Shootings)) +
  geom_line() +
  geom_point(
    data = NYPD_time_year %>% slice_max(Shootings, n = 2),
    aes(color = "Peak Days")
  ) +
  scale_x_date(date_labels = "%b") +
  labs(
    title = "NYPD Shootings by Day of Year",
    subtitle = "2006-2024",
    x = "Day",
    y = "Shootings",
    color = "",
    caption = "Figure 6 - Shootings by Day of Year"
```

```
) +
  theme_minimal()
```

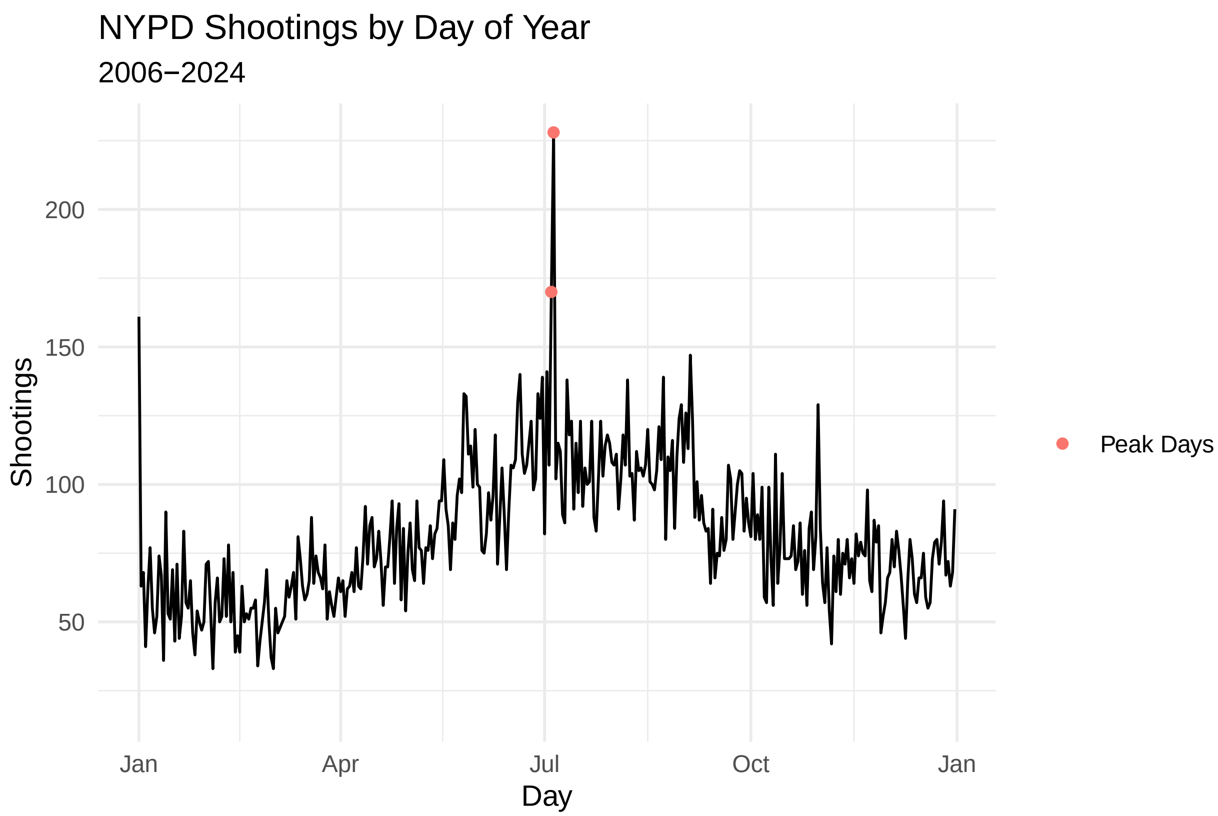

Figure 6 − Shootings by Day of Year

```
#Shootings by Time of Day

NYPD_time_day <- NYPD_clean %>%
  group_by(OCCUR_TIME) %>%
  summarize(
    Shootings = sum(Shootings),
    Murders = sum(STATISTICAL_MURDER_FLAG),
    .groups = "drop"
  )
```

```
#Shootings by Time of Day
NYPD_time_day %>%
  ggplot(aes(OCCUR_TIME, Shootings)) +
  geom_line() +
  scale_x_time() +
  labs(
    title = "NYPD Shootings by Time of Day",
    subtitle = "2006-2024",
    x = "Time (24-hour)",
    y = "Shootings",
    caption = "Figure 7 - Shootings by Time of Day"
  ) +
```

```
  theme_minimal()
```

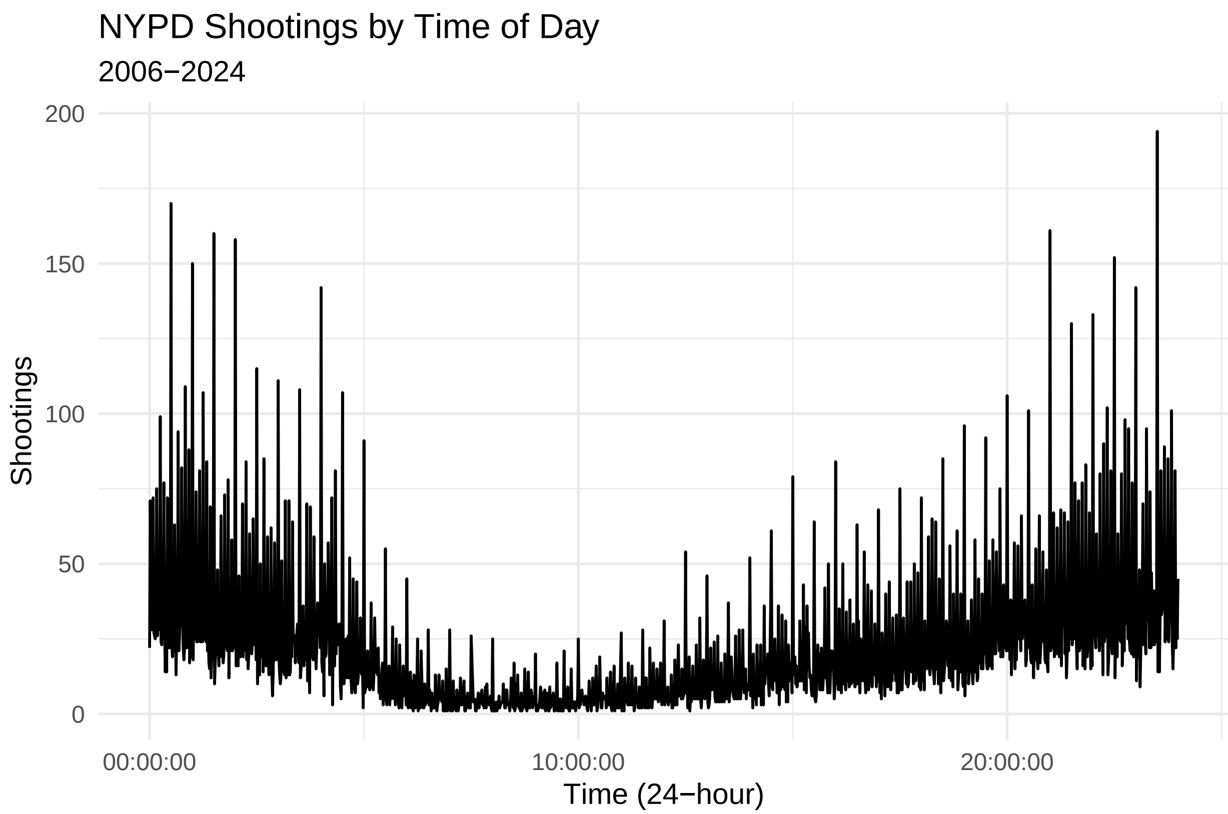

Figure 7 − Shootings by Time of Day

```
NYPD_clean %>%
  count(Dow, Hour, name = "Shootings") %>%
  ggplot(aes(Hour, Dow, fill = Shootings)) +
  geom_tile() +
  scale_fill_viridis_c() +
  labs(
    title = "Heatmap of Shootings by Day of Week and Hour",
    x = "Hour of day (24h)",
    y = "Day of week",
    fill = "Shootings",
    caption = "Figure 8"
  )
```

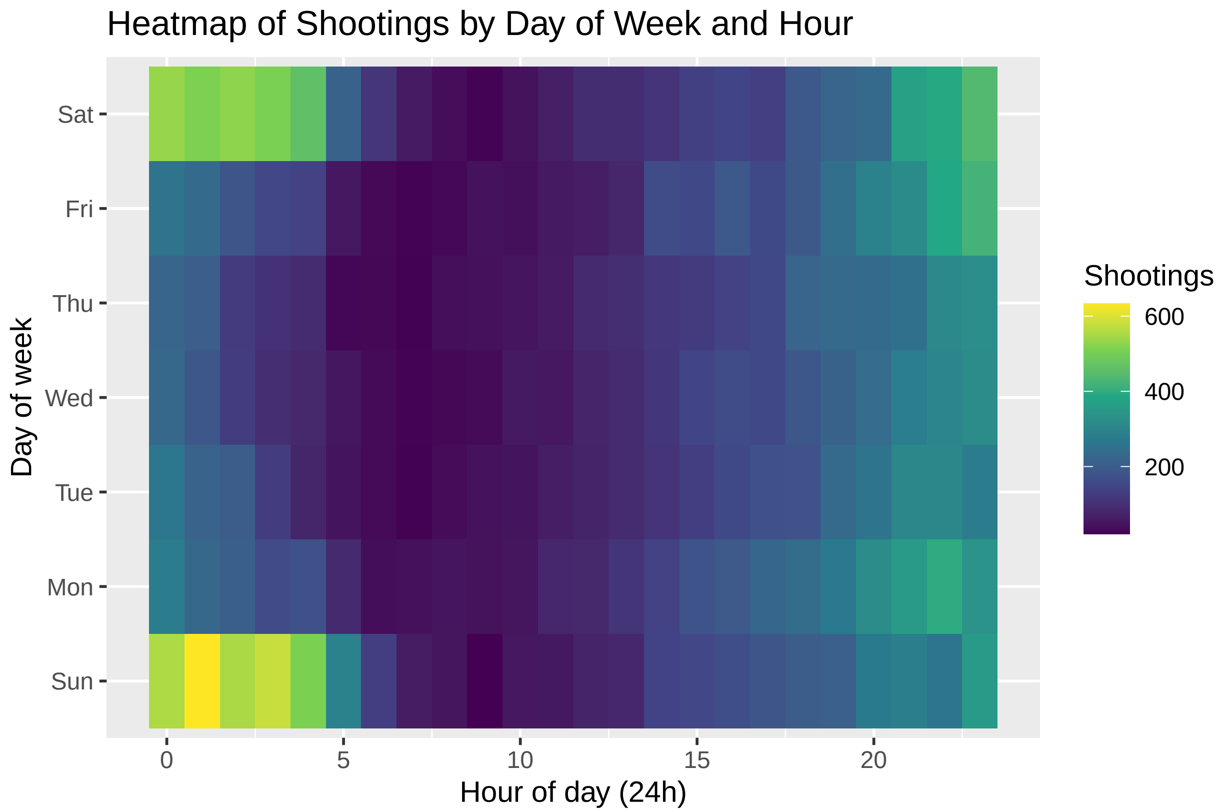

Figure 8

```r
#Victim demographics

#Gender
NYPD_clean %>%
  count(VIC_SEX, name = "Shootings") %>%
  ggplot(aes(fct_reorder(VIC_SEX, Shootings), Shootings, fill = VIC_SEX)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Shooting Incidents by Victim Sex",
    x = "Victim sex",
    y = "Number of shootings",
    caption = "Figure 9 - Gender"
  ) +
  theme(legend.position = "none")
```

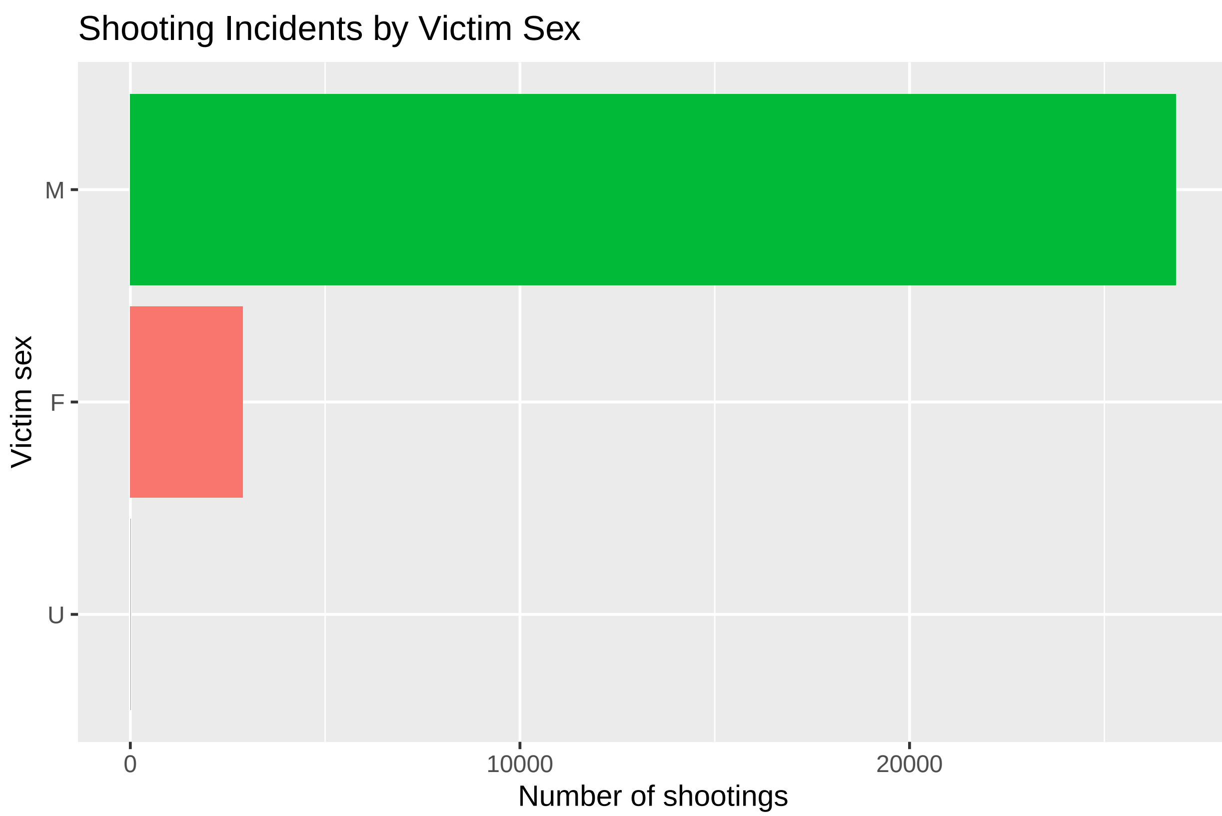

Figure 9 − Gender

```r
#Race
NYPD_clean %>%
  count(VIC_RACE, name = "Shootings") %>%
  arrange(desc(Shootings)) %>%
  ggplot(aes(fct_reorder(VIC_RACE, Shootings), Shootings, fill = VIC_RACE)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Shooting Incidents by Victim Race",
    x = "Victim race",
    y = "Number of shootings",
    caption = "Figure 10 - Race"
  ) +
  theme(legend.position = "none")
```

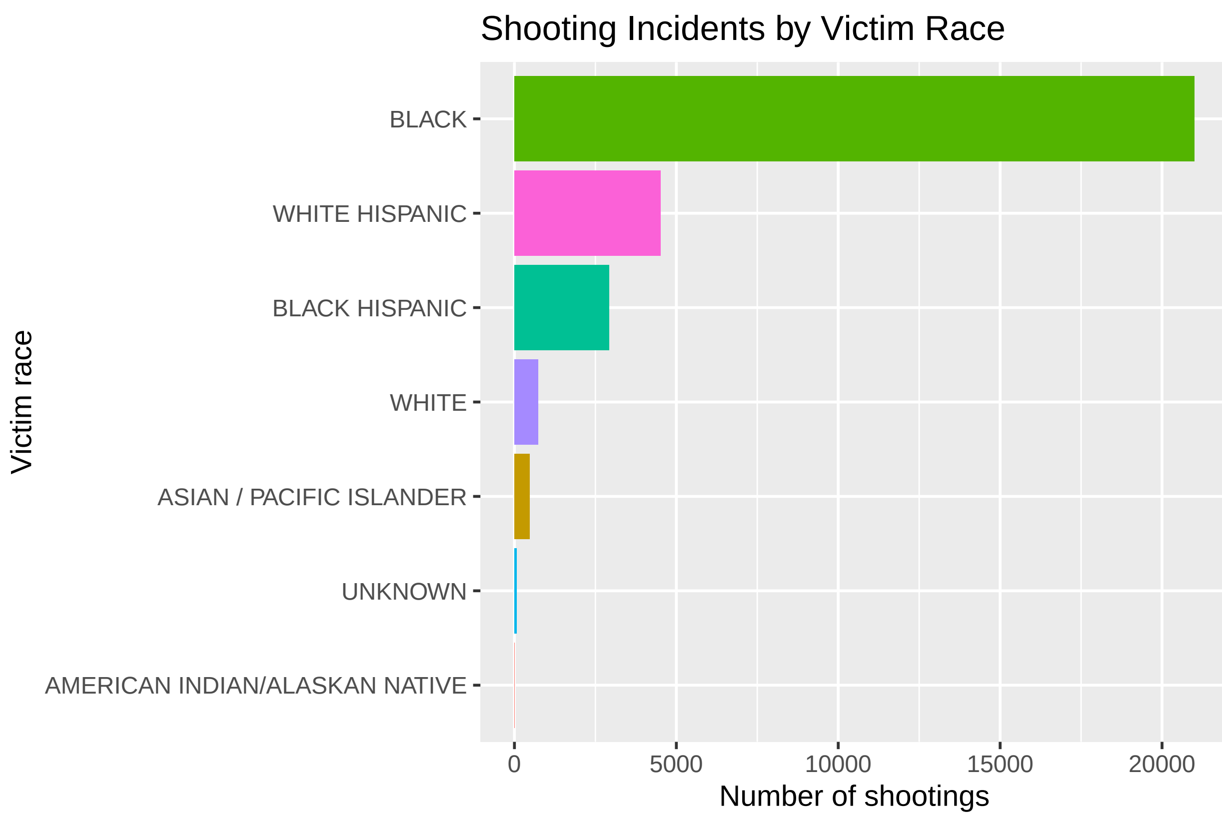

Figure 10 − Race

```r
#Age group
NYPD_clean %>%
  count(VIC_AGE_GROUP, name = "Shootings") %>%
  arrange(desc(Shootings)) %>%
  ggplot(aes(fct_reorder(VIC_AGE_GROUP, Shootings), Shootings)) +
  geom_col(fill = "darkorange") +
  coord_flip() +
  labs(
    title = "Shooting Incidents by Victim Age Group",
    x = "Victim age group",
    y = "Number of shootings",
    caption = "Figure 11 - Age"
  )
```

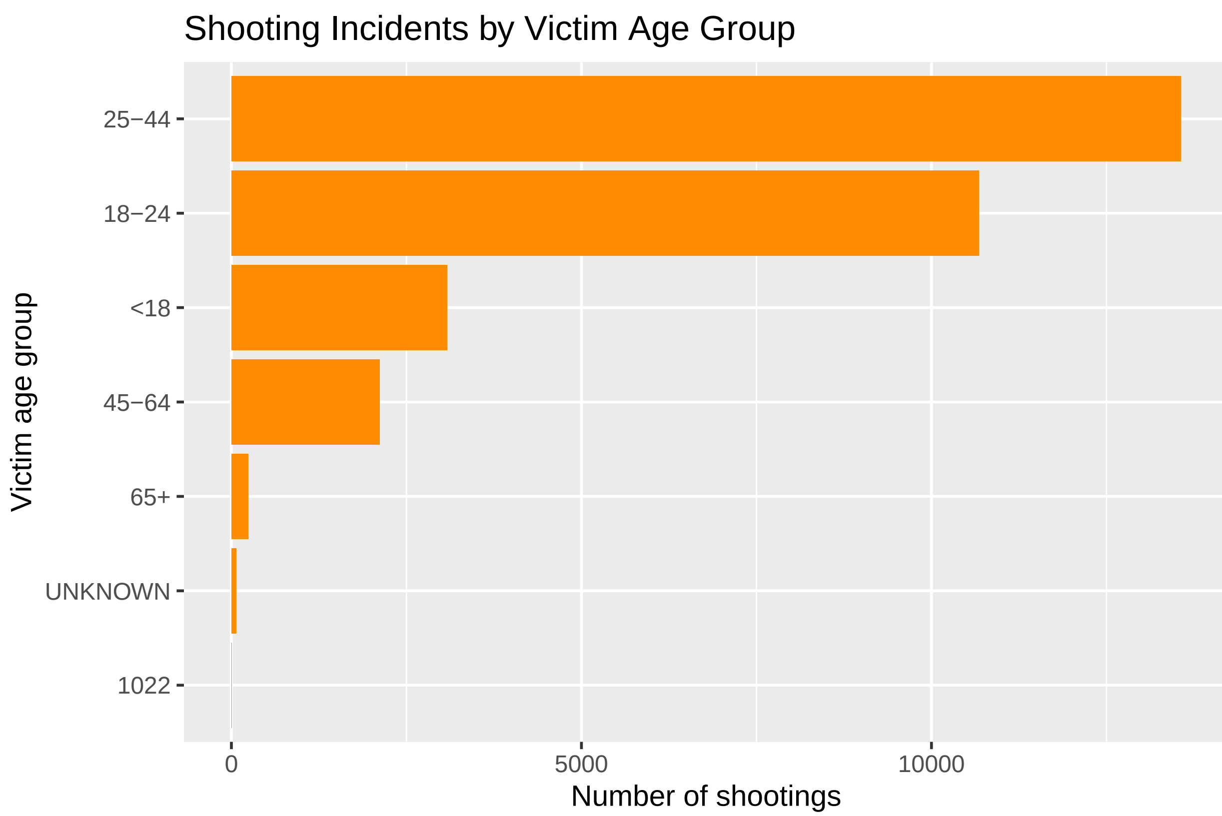

Figure 11 − Age

```r
#Model
NYPD_time_hour <- NYPD_clean %>%
  mutate(Hour = hour(OCCUR_TIME)) %>%
  group_by(Hour) %>%
  summarize(
    Shootings = sum(Shootings),
    Murders = sum(STATISTICAL_MURDER_FLAG),
    .groups = "drop"
  ) %>%
  mutate(Hour2 = Hour^2)

model <- lm(Shootings ~ Hour + Hour2, data = NYPD_time_hour)
summary(model)
```

```
##
## Call:
## lm(formula = Shootings ~ Hour + Hour2, data = NYPD_time_hour)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -443.63 -160.45   59.06  187.37  318.96
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 2368.156    141.318   16.76 1.25e-13 ***
## Hour        -352.119     28.461  -12.37 4.14e-11 ***
```

```
## Hour2         16.210      1.195   13.56 7.37e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 250.3 on 21 degrees of freedom
## Multiple R-squared:  0.9013, Adjusted R-squared:  0.8919
## F-statistic: 95.93 on 2 and 21 DF,  p-value: 2.744e-11
```

```r
#Hourly Shootings with Quadratic Fit
NYPD_time_hour %>%
  ggplot(aes(Hour, Shootings)) +
  geom_point(aes(color = "Data Points")) +
  stat_smooth(
    aes(color = "Quadratic Fit"),
    method = "lm",
    formula = y ~ x + I(x^2),
    size = 1
  ) +
  labs(
    title = "NYPD Shootings by Hour of Day",
    subtitle = "2006-2024",
    x = "Hour (24-hour)",
    y = "Shootings",
    color = "Legend",
    caption = "Figure 12"
  ) +
  theme_minimal()
```

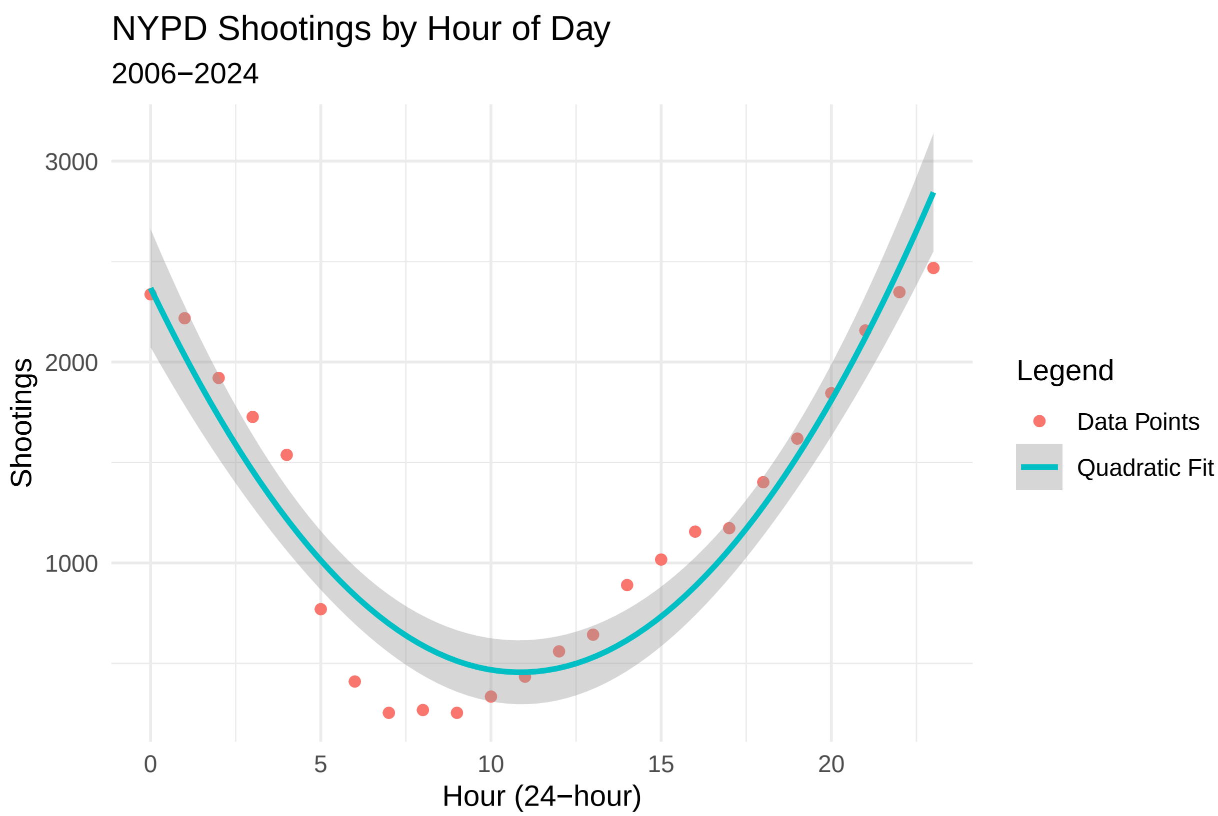
NYPD Shootings by Hour of Day
2006−2024
Shootings
3000
2000
1000
0
5
10
15
20
Hour (24−hour)
Legend
Data Points
Quadratic Fit

Figure 12